

DATE: ____/____/____

Min Stack (L 4)

SUBJECT: _____

LECTURE: _____

DATE: ____/____/____

IMPORTANT (EQUATIONS, LAWS, ETC.)

NOTES

Next Greater Element $\angle - 5$

Monotonic stack $\rightarrow$ where we store elements in a specific order either asc order

input arr : 6, 0, 8, 1, 3

Output 8 8 -1 3 -1

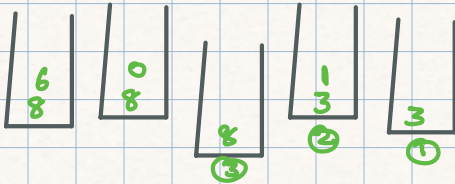
IMPORTANT (EQUATIONS, LAWS, ETC.)

Optimised : we need a stack because we don't want to get the largest but get largest w.r.t to index

NOTES

$i \leftarrow i$

6	0	8	1	3
8	8	-1	3	-1



① stack is empty

ans[i] = -1

stack.push(arr[i])

② !stack.isEmpty() &&

stack.peek > arr[i]

ans[i] = stack.peek()

stack.push(arr[i])

③ !stack.isEmpty() &&

stack.peek <= arr[i]

while (!stack.isEmpty() &&
stack.peek < arr[i]) {

stack.pop();

}

ans[i] = !stack.isEmpty() ? stack.peek() : -1